

Hayabusa2



Artist's impression of *Hayabusa2* firing its ion thrusters

Mission type	Asteroid sample-return
Operator	JAXA
COSPAR ID	2014-076A (https://nssdc.gsfc.nasa.gov/nmc/spacecraft/display.action?id=2014-076A)
SATCAT no.	40319
Website	www.hayabusa2.jaxa.jp/en/ (http://www.hayabusa2.jaxa.jp/en/)
Mission duration	6 years (planned) (11 years, 5 months and 26 days elapsed)

Spacecraft properties	
Spacecraft type	Hayabusa
Manufacturer	NEC ^[1]
Launch mass	600 kg ^[2]
Dry mass	490 kg (1,080 lb) ^[3]
Dimensions	Spacecraft bus : 1 × 1.6 × 1.25 m (3 ft 3 in × 5 ft 3 in × 4 ft 1 in) Solar panel : 6 m × 4.23 m (19.7 ft × 13.9 ft)
Power	2.6 kW (at 1 au), 1.4 kW (at 1.4 au)

Start of mission	
Launch date	3 December 2014, 04:22:04 UTC ^[4]
Rocket	H-IIA 202
Launch site	Tanegashima Space Center, LA-Y
Contractor	Mitsubishi Heavy Industries

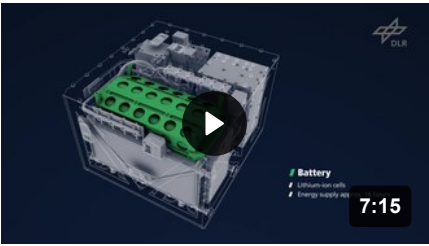
End of mission	
Landing date	Re-entry capsule: 5 December 2020 UTC^[5]
Landing site	Woomera, Australia
Flyby of Earth	
Closest approach	3 December 2015
Distance	3,090 km (1,920 mi) ^[6]
Rendezvous with (162173) Ryugu	
Arrival date	27 June 2018, 09:35 UTC ^[7]
Departure date	12 November 2019 ^[8]
Sample mass	5.4 grams ^[9] (including gas samples)
(162173) Ryugu lander	
Landing date	21 February 2019
(162173) Ryugu lander	
Landing date	11 July 2019
Flyby of Earth (Sample return)	
Closest approach	5 December 2020 UTC ^[5]

Hayabusa2 (Japanese: はやぶさ 2; lit. 'Peregrine falcon 2') is an asteroid sample-return mission operated by the Japanese state space agency JAXA. It is a successor to the *Hayabusa* mission, which returned asteroid samples for the first time in June 2010.^[10] *Hayabusa2* was launched on 3 December 2014 and rendezvoused in space with near-Earth asteroid 162173 Ryugu on 27 June 2018.^[11] It surveyed the asteroid for a year and a half and took samples. It left the asteroid in November 2019 and returned the samples to Earth on 5 December 2020 UTC.^{[8][12][13]} Its mission has now been extended through at least 2031, when it will rendezvous with the small, rapidly-rotating asteroid 1998 KY₂₆.

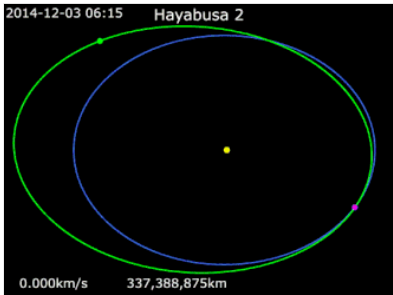
Hayabus2 carries multiple science payloads for remote sensing and sampling, and four small rovers to investigate the asteroid surface and analyze the environmental and geological context of the samples collected.

Mission overview

Asteroid 162173 Ryugu (formerly designated 1999 JU₃) is a primitive carbonaceous near-Earth asteroid. Carbonaceous asteroids are thought to preserve the most pristine, untainted materials in the Solar System, a mixture of minerals, ice, and organic compounds that interact with each other.^[14]



Hayabusa2 mission overview animation



Animation of Hayabusa2 orbit from 3 December 2014
■ Hayabusa2 ■ 162173 Ryugu ■ Earth
■ Sun
See detailed video including the extended mission

Studying it is expected to provide additional knowledge on the origin and evolution of the inner planets and, in particular, the origin of water and organic compounds on Earth,^{[14][15]} all relevant to the origin of life on Earth.^[16]

Initially, launch was planned for 30 November 2014,^{[17][18][19]} but was delayed to 3 December 2014 at 04:22:04 UTC (3 December 2014, 13:22:04 local time) on a H-IIA launch vehicle.^[20] *Hayabusa2* launched together with PROCYON asteroid flyby space probe. PROCYON's mission was a failure. *Hayabusa2* arrived at Ryugu on 27 June 2018,^[11] where it surveyed the asteroid for a year and a half and collected samples.^[14] It departed the asteroid in November 2019 and returned the samples to Earth in December 2020.^[19]

Compared to the previous *Hayabusa* mission, the spacecraft features improved ion engines, guidance and navigation technology, antennas, and attitude control systems.^[21] A kinetic penetrator (a high-explosive shaped charge) was shot into the asteroid surface to expose pristine sample material which was later collected for return to Earth.^{[15][19]}

Funding and history

Following the initial success of *Hayabusa*, JAXA began studying a potential successor mission in 2007.^[22] In July 2009, Makoto Yoshikawa of JAXA presented a proposal titled "Hayabusa Follow-on Asteroid Sample Return Missions". In August 2010, JAXA obtained approval from the Japanese government to begin development of *Hayabusa2*. The cost of the project estimated in 2010 was 16.4 billion yen (US\$149 million).^{[10][23]}

Hayabusa2 was launched on 3 December 2014, arrived at asteroid Ryugu on 27 June 2018, and remained stationary at a distance of about 20 km (12 mi) to study and map the asteroid. In the week of 16 July 2018, commands were sent to move to a lower hovering altitude.^[24]

On 21 September 2018, the *Hayabusa2* spacecraft ejected the first two rovers, Rover-1A (HIBOU)^[25] and Rover-1B (OWL), from about a 55 m (180 ft) altitude that dropped independently to the surface of the asteroid.^{[26][27]} They functioned nominally and transmitted data.^[28] The MASCOT rover deployed successfully on 3 October 2018 and operated for about 16 hours as planned.^[29]

The first sample collection was scheduled to start in late October 2018, but the rovers encountered a landscape with large and small boulders but no surface soil for sampling. Therefore, it was decided to postpone the sample collection plans to 2019 and further evaluate various options for the landing.^{[30][31]} The first surface sample retrieval took place on 21 February 2019. On 5 April 2019, *Hayabusa2* released an impactor to create an artificial crater on the asteroid surface. However, *Hayabusa2* initially failed on 14 May 2019 to drop reflective markers necessary onto the surface for guiding the descent and sampling processes,^[32] but later it successfully dropped one from an altitude of 9 m (30 ft) on 4 June 2019.^[33] The sub-surface sampling took place on 11 July 2019.^[34] The spacecraft departed the asteroid on 13 November 2019 (with departure command sent at 01:05 UTC on 13 November 2019). It successfully delivered the samples back to Earth on 6 December 2020 (JST), dropping the contents by parachute in a special container at a location in southern Australia. The samples were retrieved the same day for secure transport back to the JAXA labs in Japan.^{[8][35][36]}

Spacecraft

The design of *Hayabusa2* is based on the first *Hayabusa* spacecraft, with some improvements.^{[14][39]} It has a mass of 600 kilograms (1,300 lb) including fuel,^[39] and electric power is generated by two sets of solar arrays with an output of 2.6 kW at 1 AU, and 1.4 kW at 1.4 AU.^[39] The power is stored in eleven inline-mounted 13.2 Ah lithium-ion batteries.^[39]

Propulsion

The spacecraft features four solar-electric ion thrusters for propulsion called μ 10,^[37] one of which is a backup. These engines use microwaves to convert xenon into plasma (ions), which are accelerated by a voltage applied by the solar panels and ejected out the back of the engine. The simultaneous operation of three engines generates thrusts of up to 28 mN.^[39] Although this thrust is very small, the engines are also extremely efficient; the 66 kg (146 lb) of xenon^[37] reaction mass can change the speed of the spacecraft by up to 2 km/s.^[39]

The spacecraft has four redundant reaction wheels and a chemical reaction control system featuring twelve thrusters for attitude control (orientation) and orbital control at the asteroid.^{[37][39]} The chemical thrusters use hydrazine and MON-3, with a total mass of 48 kg (106 lb) of chemical propellant.^[39]

Communication

<i>Hayabusa2</i>	Performance ^{[37][38]}
Propulsion	μ 10 ion thruster
Number of thrusters	4 (one is a spare)
Total thrust (ion drive)	28 mN
Specific impulse (<i>I</i> _{sp})	3000 seconds
Acceleration	49 μ m/s ²
Power	1250 W
Spacecraft wet mass	600 kg
Ion engine system dry mass	66 kg
Ion engine system wet mass	155 kg
Solar array	23 kg
Xenon propellant	66 kg
Hydrazine/MON-3 propellant	48 kg
Thrust (chemical propellants)	20 N

The primary contractor NEC built the 590 kg (1,300 lb) spacecraft, its Ka-band communications system and a mid-infrared camera.^[21] The spacecraft has two high-gain directional antennas for X-band and K_a-band.^[37] Bit rates are 8 bit/s to 32 kbit/s.^[39] The ground stations are the Usuda Deep Space Center, Uchinoura Space Center, NASA Deep Space Network and Malargüe Station (ESA).^[39]

Navigation

The optical navigation camera telescope (ONC-T) is a telescopic framing camera with seven colors to optically navigate the spacecraft.^[40] It works in synergy with the optical navigation camera wide-field (ONC-W2) and with two star trackers.^[39]

In order to descend to the asteroid surface to perform sampling, the spacecraft released one of five target markers in the selected landing zones as artificial guide marks, with highly reflective outer material that is recognized by a strobe light mounted on the spacecraft.^[39] The spacecraft also used its laser altimeter and ranging (LIDAR) as well as Ground Control Point Navigation (GCP-NAV) sensors during sampling.^[39]

Firsts

The *Hayabusa2* spacecraft was the first to deploy operating rovers on an asteroid.

Science payload

The *Hayabusa2* payload is equipped with multiple scientific instruments:^{[39][41]}

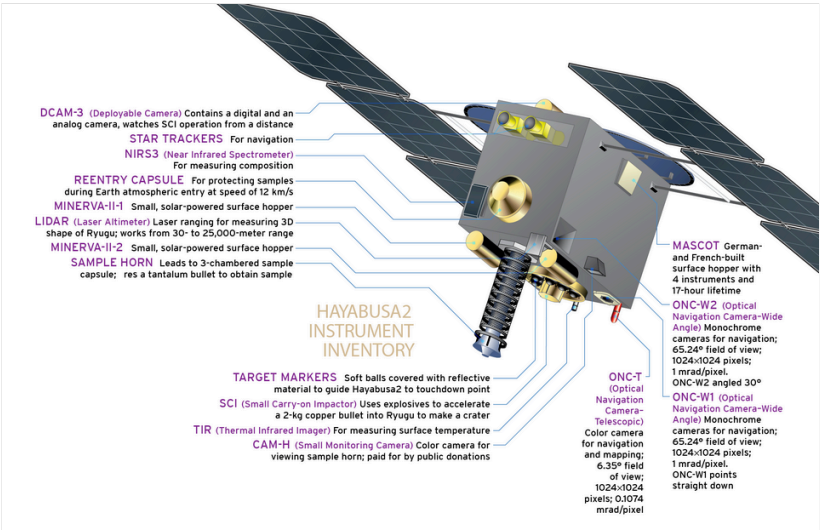
- Remote sensing: Optical Navigation Camera (ONC-T, ONC-W1, ONC-W2), Near-Infrared Camera (NIR3), Thermal-Infrared Camera (TIR), Light Detection And Ranging (LIDAR)
- Sampling: Sampling device (SMP), Small Carry-on Impactor (SCI), Deployable Camera (DCAM3)
- Four rovers: Mobile Asteroid Surface Scout (MASCOT), Rover-1A, Rover-1B, Rover-2.

Remote sensing

The Optical Navigation Cameras (ONCs) were used for spacecraft navigation during the asteroid approach and proximity operations. They also remotely imaged the surface to search for interplanetary dust around the asteroid. ONC-T is a telephoto camera with a 6.35° × 6.35° field of view and several optical filters carried in a carousel. ONC-W1 and ONC-W2 are wide angle (65.24° × 65.24°) panchromatic (485–655 nm) cameras with nadir and oblique views, respectively.^[39]

The Near-Infrared Spectrometer (NIRS3) is a spectrograph operating at a wavelength of 1.8–3.2 μm. NIRS3 was used for analysis of surface mineral composition.^[39]

The Thermal-Infrared Imager (TIR) is a thermal infrared camera working at 8–12 μm, using a two-dimensional microbolometer array. Its spatial resolution is 20 m at 20 km distance or 5 cm at 50 m distance (70 ft at 12 mi, or 2 in at 160 ft). It was used to determine surface temperatures in the range –40 to 150 °C (–40 to 302 °F).^[39]



Hayabusa2 instrument inventory

The Light Detection And Ranging (LIDAR) instrument measured the distance from the spacecraft to the asteroid surface by measuring the reflected laser light. It operated over an altitude range between 30 m and 25 km (100 ft and 16 mi).^[39]

When the spacecraft was closer to the surface than 30 m (98 ft) during the sampling operation, the Laser Range Finders (LRF-S1, LRF-S3) were used to measure the distance and the attitude (orientation) of the spacecraft relative to the terrain.^{[42][43]} The LRF-S2 monitored the sampling horn to trigger the sampling projectile.

LIDAR and ONC data are being combined to determine the detailed topography (dimensions and shape) of the asteroid. Monitoring of a radio signal from Earth allowed measurement of the asteroid's gravitational field.^[39]

Rovers

Hayabusa2 carried four small rovers to explore the asteroid surface *in situ*,^[44] and provide context information for the returned samples. Due to the minimal gravity of the asteroid, all four rovers were designed to move around by short hops instead of using normal wheels. They were deployed at different dates from about 60 m (200 ft) altitude and fell freely to the surface under the asteroid's weak gravity.^[45] The first two rovers, called HIBOU (previously Rover-1A) and OWL (previously Rover-1B), landed on asteroid Ryugu on 21 September 2018.^[28] The third rover, called MASCOT, was deployed 3 October 2018. Its mission was successful.^[46] The fourth rover, known as Rover-2 or

MINERVA-II-2, failed before release from the orbiter. It was released on 2 October 2019 to orbit the asteroid and perform gravitational measurements before being allowed to impact the asteroid a few days later.

MINERVA-II

MINERVA-II is a successor to the MINERVA lander carried by Hayabusa. It consists of two containers with 3 rovers.

MINERVA-II-1 is a container that deployed two rovers, **Rover-1A (HIBOU)** and **Rover-1B (OWL)**, on 21 September 2018.^{[47][48]} It was developed by JAXA and the University of Aizu. The rovers are identical having a cylindrical shape, 18 cm (7.1 in) diameter and 7 cm (2.8 in) tall, and a mass of 1.1 kg (2.4 lb) each.^{[39][49]} They move by hopping in the low gravitational field, using a torque generated by rotating masses within the rovers.^[50] Their scientific payload is a stereo camera, wide-angle camera, and thermometers. Solar cells and double-layer capacitors provide the electrical power.^{[2][51]} The MINERVA-II-1 rovers were successfully deployed 21 September 2018. Both rovers performed successfully on the asteroid surface, sending images and video from the surface. Rover-1A operated for 113 asteroid days (36 Earth days) returning 609 images from the surface, and Rover-1B operated for 10 asteroid days (3 Earth days) returning 39 images from the surface.^[52]

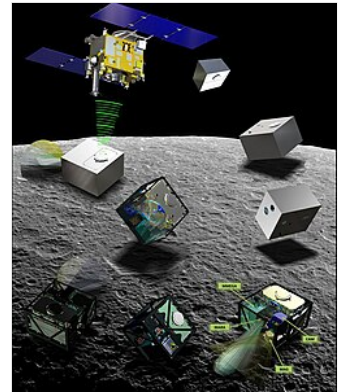
The MINERVA-II-2 container held the **ROVER-2** (sometimes referred to as MINERVA-II-2), developed by a consortium of universities led by Tohoku University in Japan. This was an octagonal prism shape, 15 cm (5.9 in) diameter and 16 cm (6.3 in) tall, with a mass of about 1 kg (2.2 lb). It had two cameras, a thermometer and an accelerometer. It was equipped with optical and ultraviolet LEDs to illuminate and detect floating dust particles. ROVER-2 carried four mechanisms to move around using short hops.^[2] Rover-2 had problems prior to deployment from the orbiter but was released on 2 October 2019 to orbit the asteroid and perform gravitational measurements. It was then crashed onto the asteroid surface a few days later on 8 October 2019.

MASCOT

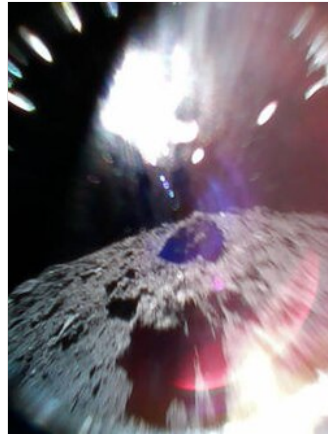
The **Mobile Asteroid Surface Scout (MASCOT)** was developed by the German Aerospace Center (DLR) in cooperation with the French space agency CNES.^[53] Its dimensions measure 29.5 cm × 27.5 cm × 19.5 cm (11.6 in × 10.8 in × 7.7 in) and it weighs 9.6 kg (21 lb).^[54] MASCOT carries four

instruments: an infrared spectrometer (MicrOmega), a magnetometer (MASMAG), a radiometer (MARA), and a camera (MASCAM) that imaged the small-scale structure, distribution and texture of the regolith.^[55] The rover is capable of tumbling once to reposition itself for further measurements.^{[44][56]} It collected data on the surface structure and mineralogical composition, the thermal behaviour and the magnetic properties of the asteroid.^[57] It has a non-rechargeable battery that allowed for operations for approximately 16 hours.^{[58][59]} The infrared radiometer on the *InSight* Mars lander, launched in 2018, is based on the MASCOT radiometer.^{[60][61]}

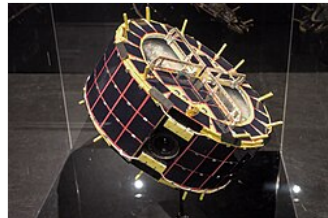
MASCOT was deployed on 3 October 2018, with a successful landing and ensuing surface mission. Two papers were published describing the results from MASCOT in the scientific journals *Nature Astronomy*^[62] and *Science*.^[63] One finding of the research was that C-type asteroids consist of more porous material than previously thought, explaining a deficit of this meteorite type. Meteorites of this type are too porous to survive the entry into the atmosphere of planet Earth. Another finding was that Ryugu consists of two different almost black types of rock with little internal cohesion, but no dust was detected.^{[64][65]} A third paper describing results from MASCOT was published in the *Journal of Geophysical Research* and describes the magnetic properties of Ryugu, showing that Ryugu does not have a magnetic field on a boulder scale.^[66]



Mission overview



The first photograph from the surface of an asteroid, taken by *HIBOU* on 22 September 2018 during one of its "hops".



Model of MINERVA-II-1 Rover 1B

Objects deployed by Hayabusa2

Object	Developed by	Mass	Dimensions	Power	Science payload	Landing or deployed date	Status
MINERVA-II-1 rovers: Rover-1A (HIBOU) Rover-1B (OWL)	JAXA and University of Aizu	1.1 kg (2.4 lb) each	Diameter: 18 cm (7.1 in) Height: 7 cm (2.8 in)	Solar panels	Wide-angle camera, stereo camera, thermometers	21 September 2018	Successful landing. Rover-1A operated for 36 days and Rover-1B operated for 3 days. ^[52]
Rover-2 (MINERVA-II-2)	Tohoku University	1.0 kg (2.2 lb)	Diameter: 15 cm (5.9 in) Height: 16 cm (6.3 in)	Solar panels	Two cameras, thermometer, accelerometer. Optical and ultraviolet LEDs for illumination	Released: 2 October 2019, 16:38 UTC	Rover failed before deployment, so it was released in orbit around the asteroid to perform gravitational measurements before it impacted a few days later. ^{[67][68]}
MASCOT	German Aerospace Center and CNES	9.6 kg (21 lb)	29.5 cm × 27.5 cm × 19.5 cm (11.6 in × 10.8 in × 7.7 in)	Non-rechargeable battery ^[58]	Camera, infrared spectrometer, magnetometer, radiometer	3 October 2018 ^[69]	Successful landing. Operated on battery for more than 17 hours ^[59]
Deployable camera 3 (DCAM3)	JAXA	about 2 kg (4.4 lb)	Diameter: 7.8 cm (3.1 in) Height: 7.8 cm (3.1 in)	Non-rechargeable battery	DCAM3-A lens, DCAM3-D lens	5 April 2019	Deployed to observe impact of SCI impactor. Inactive now and presumed to have fallen on the asteroid.
Small Carry-On Impactor (SCI)	JAXA	2.5 kg (5.5 lb)	Diameter: 30 cm (12 in) Height: 21.7 cm (8.5 in)	Non-rechargeable battery	None	5 April 2019	Successful. Shot to the surface 40 minutes after separation.
Target Marker B	JAXA	300 g (11 oz)	10 cm (3.9 in) sphere	None	None	25 October 2018	Successful. Used for first touchdown.
Target Marker A	JAXA	300 g (11 oz)	10 cm (3.9 in) sphere	None	None	30 May 2019	Successful. Used for second touchdown.

Target Marker E (Explorer)	JAXA	300 g (11 oz)	10 cm (3.9 in) sphere	None	None	17 September 2019	Successful. Injected to equatorial orbit and confirmed to land.
Target Marker C (Sputnik/Спутник)	JAXA	300 g (11 oz)	10 cm (3.9 in) sphere	None	None	17 September 2019	Successful. Injected to polar orbit and confirmed to land.
Target Marker D	JAXA	300 g (11 oz)	10 cm (3.9 in) sphere	None	None	—	Was not deployed.
Sample Return Capsule	JAXA	16 kg	Diameter: 40 cm Height: 20 cm	Non-rechargeable battery	Sample container, Reentry flight Environment Measurement Module	5 December 2020 UTC	Successful landing. All the parts including the sample container were collected.

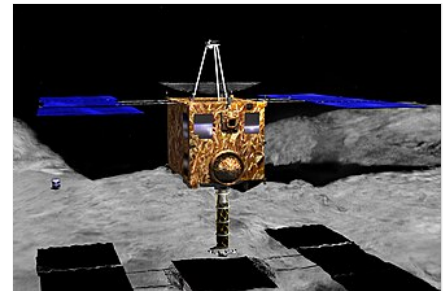
Sampling

The original plan was for the spacecraft to collect up to three samples: 1) surface material that exhibits traits of hydrous minerals; 2) surface material with either unobservable or weak evidence of aqueous alterations; 3) excavated sub-surface material.^[71]

The first two surface samples were scheduled to start in late October 2018, but the rovers showed large and small boulders and insufficient surface area to sample, so the mission team decided to postpone sampling to 2019 and evaluate various options.^[30] The first surface sampling was completed on 22 February 2019 and obtained a substantial amount of topsoil,^{[70][72]} so the second surface sampling was postponed and was eventually cancelled to decrease the risks to the mission.^[70]

The second and final sample was collected from material that was dislodged from beneath the surface by the kinetic impactor (SCI impactor) shot from a distance of 300 m (980 ft).^{[73][74]} All samples are stored in separate sealed containers inside the sample return capsule (SRC).

Sampling	Date
1st surface sampling	21 February 2019
Sub-surface sampling	SCI impactor: 5 April 2019 Target marker: 5 June 2019 ^[33] Sampling: 11 July 2019 ^[34]
2nd surface sampling	Optional; ^[70] was not done.



Artistic rendering of Hayabusa collecting a surface sample.

Surface sample

Hayabus2's sampling device is based on *Hayabusa's*. The first surface sample retrieval was conducted on 21 February 2019, which began with the spacecraft's descent, approaching the surface of the asteroid. When the sampler horn attached to *Hayabus2's* underside touched the surface, a 5 g (0.18 oz) tantalum projectile (bullet) was fired at 300 m/s (980 ft/s) into the surface.^[72] The resulting ejected materials were collected by a "catcher" at the top of the horn, which the ejecta reached under their own momentum under microgravity conditions.^[75]



Sampler horn

Sub-surface sample



Animation illustrating SCI deployment and subsequent sampling from the resulting crater.

The sub-surface sample collection required an impactor to create a crater in order to retrieve material under the surface, not subjected to space weathering. This required removing a large volume of surface material with a powerful impactor. For this purpose, *Hayabus2* deployed on 5 April 2019 a free-flying gun with one "bullet", called the **Small Carry-on Impactor (SCI)**; the system contained a 2.5 kg (5.5 lb) copper projectile, shot onto the surface with an explosive propellant charge. Following SCI deployment, *Hayabus2* also left behind a deployable camera (**DCAM3**)^[Note 1] to observe and map the precise location of the SCI impact, while the orbiter maneuvered to the far side of the asteroid to avoid being hit by debris from the impact.

It was expected that the SCI deployment would induce seismic shaking of the asteroid, a process considered important in the resurfacing of small airless bodies. However, post-impact images from the spacecraft revealed that little shaking had occurred, indicating the asteroid was significantly less cohesive than was expected.^[76]

Approximately 40 minutes after separation, when the spacecraft was at a safe distance, the impactor was fired into the asteroid surface by detonating a 4.5 kg (9.9 lb) shaped charge of plasticized HMX for acceleration.^{[56][77]} The copper impactor was shot onto the surface from an altitude of about 500 m (1,600 ft) and it excavated a crater of about 10 m (33 ft) in diameter, exposing pristine material.^{[15][32]} The next step was the deployment on 4 June 2019 of a reflective target marker in the area near the crater to assist with navigation and descent.^[33] The touchdown and sampling took place on 11 July 2019.^[34]

Sample return

The spacecraft collected and stored the samples in separate sealed containers inside the sample-return capsule (SRC), which is equipped with thermal insulation. The container is 40 cm (16 in) external diameter, 20 cm (7.9 in) in height, and a mass of about 16 kg (35 lb).^[39]

At the end of the science phase in November 2019,^[8] *Hayabus2* used its ion engines to change orbit and return to Earth.^[75] Hours before *Hayabus2* flew past Earth in late 2020, it released the capsule, on 5 December 2020 at 05:30 UTC.^[78] The capsule was released spinning at one revolution per three seconds, and re-entered the Earth's atmosphere at 12 km/s (7.5 mi/s). Deploying a radar-reflective parachute at an altitude of about 10 km (6.2 mi), it ejected its heat shield while transmitting a position beacon signal.^{[39][75]} The sample capsule landed at the Woomera Test Range in Australia.^{[13][79]} The total flight distance was 5.24×10^9 km (35.0 AU).^[39]

Any volatile substances will be collected before the sealed containers are opened.^[71] The samples will be curated and analyzed at JAXA's Extraterrestrial Sample Curation Center,^[80] where international scientists can request a small portion of the samples. The spacecraft brought back a capsule containing carbon-rich asteroid fragments that scientists believe could provide clues about the ancient delivery of water and organic molecules to Earth.^{[81][82]}

JAXA is sharing a portion of these samples with NASA. In exchange, NASA will provide JAXA a percentage of a sample of asteroid Bennu, when the agency's OSIRIS-REx spacecraft returned to Earth from the space rock on 24 September 2023.^[83]

A November 2024 study published in *Meteoritics & Planetary Science* examined a sample returned from the asteroid Ryugu by the Hayabus2 mission. Despite stringent contamination control measures, terrestrial microorganisms rapidly colonized the sample. Researchers concluded that the observed microbial growth originated from Earthly contamination during sample preparation rather than being



The touchdown on and sampling of Ryugu on 11 July



Replica of *Hayabusa's* sample-return capsule (SRC) used for re-entry. *Hayabus2's* capsule is of the same size, measuring 40 cm (16 in) in diameter and using a parachute for touchdown.



One of the facility-to-facility transfer containers (FFTC) of Hayabus2 returned samples given to NASA by JAXA.

indigenous to the asteroid. This finding underscores the difficulty of preventing microbial contamination, even in highly controlled laboratory environments.^{[84][85]}

Mission extension (Hayabusa2#)

With the successful return and retrieval of the sample capsule on 6 December 2020 (JST), Hayabusa2 is planned to use its remaining 30 kg (66 lb) of xenon propellant (from the initial 66 kg (146 lb)) to extend its service life and fly out to explore new targets.^[86] As of September 2020, a fly-by of 98943 Torifune in July 2026 and a rendezvous with 1998 KY₂₆ in July 2031 were selected for the mission extension.^{[87][88][89]} The observation of Torifune will be a high-speed fly-by of an S-type asteroid.^[90] The fixed camera of Hayabusa2 was not designed for this type of fly-by. The rendezvous with 1998 KY₂₆ will be the first visit of a fast rotating micro-asteroid, with a rotation period of about 10 minutes.^[89] Between 2021 and 2026, the spacecraft will also conduct transit observations of exoplanets.^[89] An option to conduct a Venus flyby to set up an encounter with 2001 AV₄₃ was also studied.^{[91][92]}

Selected EAEEA (Earth → Asteroid → Earth → Earth → Asteroid) scenario:^[89]

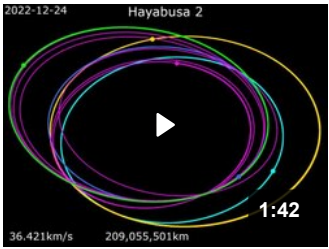
- December 2020: Extension mission start
- 2021 until July 2026: cruise operation
- July 2026: S-type asteroid 98943 Torifune high-speed fly-by
- December 2027: Earth swing-by
- June 2028: Second Earth swing-by
- July 2031: Target body (1998 KY₂₆) rendezvous

Dark Comet

The 2017 discovery of interstellar visitor ‘Oumuamua and its unusual acceleration led to the discovery of a new class of objects called dark comets.^[93] By 2024, 14 previously discovered asteroids had been identified as dark comets, including 1998 KY₂₆. At the time, Hayabusa2 had already been set on course to visit 1998 KY₂₆.^[93] Researchers who study dark comets are now very excited about the prospect of Hayabusa2#’s upcoming visit to a dark comet.^[93]

See also

- Abiogenesis
- Hayabusa Mk2 – Proposed Japanese space mission



Animation of Hayabusa2 orbit – extended mission
Hayabusa2 · 162173 Ryugu · Earth · Sun · 98943 Torifune · 1998 KY₂₆

- OSIRIS-REx – NASA asteroid sample return mission to 101955 Benu (operational at the same time as Hayabusa2)
- Panspermia

Japanese minor body probes

- Hiten (spacecraft) – 1990 Japanese lunar probe
- Martian Moons eXploration – Planned sample-return mission by Japan to Phobos
- OKEANOS – Proposed space probe to Trojan asteroids
- Suisei spacecraft

Notes

- DCAM3 is numbered as such because it is a follow-on to the DCAM1 and DCAM2 used for the IKAROS interplanetary solar sail

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